

Safety Data Sheet

1. PRODUCT AND COMPANY IDENTIFICATION

Product Name: 3,4-Dichloro-1-butene
Compound ID: AG003IFV
CAS Number: 760-23-6

Identified uses: Laboratory chemicals, manufacture of chemical compounds

Company: Angene

2. HAZARDS IDENTIFICATION

GHS Classification in accordance with 29 CFR 1910 (OSHA HCS)

Flammable liquids,(Category 3), H226
Skin corrosion/irritation,(Category 1A, 1B, 1C), H314
Hazardous to the aquatic environment, acute hazard,(Category 3), H402
Acute toxicity, oral; acute toxicity, inhalation,(Category 4), H302+H332
For the full text of the H-Statements mentioned in this Section, see Section 16.

Pictogram



Signal Word danger

Hazard statements

H226 Flammable liquid and vapor
H314 Causes severe skin burns and eye damage
H402 Harmful to aquatic life
H302+H332 Harmful if swallowed or if inhaled

Precautionary statements

P233 Keep container tightly closed.
P240 Ground/bond container and receiving equipment.
P260 Do not breathe dust/fume/gas/mist/vapours/spray.
P264 Wash hands thoroughly after handling
P270 Do not eat, drink or smoke when using this product.
P271 Use only outdoors or in a well-ventilated area.
P273 Avoid release to the environment.
P280 Wear protective gloves/protective clothing/eye protection/face protection.
P405 Store locked up.
P403+P235 Store in a well-ventilated place. Keep cool.
P501 Dispose of contents/container to in accordance with local regulation

Hazards not otherwise classified (HNOC) or not covered by GHS - none

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical Name: 3,4-Dichloro-1-butene
CAS Number: 760-23-6
Molecular Formula: C₄H₆Cl₂
Molecular Weight: 124.9964 g/mol

4. FIRST AID MEASURES

Description of first aid measures

General advice

Consult a physician. Show this safety data sheet to the doctor in attendance. Move out of dangerous area.

If inhaled

If breathed in, move person into fresh air. If not breathing, give artificial respiration. Consult a physician.

In case of skin contact

Wash off with soap and plenty of water. Consult a physician.

In case of eye contact

Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.

If swallowed

Never give anything by mouth to an unconscious person. Rinse mouth with water. Consult a physician.

Most important symptoms and effects, both acute and delayed

The most important known symptoms and effects are described in the labelling (see section 2 or section 11)

Indication of any immediate medical attention and special treatment needed

no data available

5. FIREFIGHTING MEASURES

Extinguishing media Suitable extinguishing media

Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

Special hazards arising from the substance or mixture

Carbon oxides, nitrogen oxides (NOx), Hydrogen bromide gas

Advice for firefighters

Wear self contained breathing apparatus for fire fighting if necessary.

Further information

no data available

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures

Use personal protective equipment. Avoid dust formation. Avoid breathing vapours, mist or gas. Ensure adequate ventilation. Evacuate personnel to safe areas. Avoid breathing dust. For personal protection see section 8.

Environmental precautions

Do not let product enter drains.

Methods and materials for containment and cleaning up

Pick up and arrange disposal without creating dust. Sweep up and shovel. Keep in suitable, closed containers for disposal.

Reference to other sections

For disposal see section 13.

7. HANDLING AND STORAGE

Precautions for safe handling

Avoid contact with skin and eyes. Avoid formation of dust and aerosols. Provide appropriate exhaust ventilation at places where dust is formed. Normal measures for preventive fire protection. For precautions see section 2.

Conditions for safe storage, including any incompatibilities

Inert atmosphere.
2-8°C.

Specific end use(s)

Apart from the uses mentioned in section 1, no other specific uses are stipulated

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Contains no substances with occupational exposure limit values.

Appropriate engineering controls

Handle in accordance with good industrial hygiene and safety practice. Wash hands before breaks and at the end of workday.

Personal protective equipment

Eye/face protection

Face shield and safety glasses Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).

Skin protection

Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands.

Body Protection

Complete suit protecting against chemicals, The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

Respiratory protection

For nuisance exposures use type P95 (US) or type P1 (EU EN 143) particle respirator. For higher level protection use type OV/AG/P99 (US) or type ABEK-P2 (EU EN 143) respirator cartridges. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).

Control of environmental exposure

Do not let product enter drains.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance / Form:	solid
Odor:	no data available
Odor Threshold:	no data available
pH:	no data available
Melting point:	−61 °C(lit.)
Boiling point/range:	116°C at 760 mmHg
Flash point:	83 °F
Evaporation rate:	no data available
Flammability:	no data available
Upper/lower flammability:	no data available
explosive limits:	no data available
Vapor pressure:	no data available
Vapour density:	no data available
Relative density:	no data available
Water solubility:	no data available
Partition coefficient:	no data available
Auto-ignition temperature:	no data available
Decomposition Temp:	no data available
log Pow:	no data available
Viscosity:	no data available
Explosive properties:	no data available
Oxidizing properties:	no data available

Other safety information no data available

10. STABILITY AND REACTIVITY

Reactivity:	no data available
Chemical stability:	Stable under recommended storage conditions.
Possibility of hazardous reactions	no data available
Conditions to avoid	no data available
Incompatible materials	no data available
Hazardous decomposition products	no data available
Other decomposition products:	no data available
In the event of fire:	see section 5

11. TOXICOLOGICAL INFORMATION

Acute toxicity:	Classified based on available data. For more details, see section 2
Skin corrosion/irritation:	Classified based on available data. For more details, see section 2
Serious eye damage/irritation	Classified based on available data. For more details, see section 2
Respiratory or skin sensitisation	Classified based on available data. For more details, see section 2
Germ cell mutagenicity	Classified based on available data. For more details, see section 2
Carcinogenicity:	Classified based on available data. For more details, see section 2

IARC:	No component of this product present at levels greater than or equal to 0.1% is identified as probable, possible or confirmed human carcinogen by IARC.
ACGIH:	No component of this product present at levels greater than or equal to 0.1% is identified as a carcinogen or potential carcinogen by ACGIH.
NTP:	No component of this product present at levels greater than or equal to 0.1% is identified as a known or anticipated carcinogen by NTP.
OSHA:	No component of this product present at levels greater than or equal to 0.1% is identified as a carcinogen or potential carcinogen by OSHA.
Reproductive toxicity	
Specific target organ toxicity - single exposure	no data available
Specific target organ toxicity - repeated exposure	no data available
Aspiration hazard	no data available
Additional Information	no data available
To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.	

12. ECOLOGICAL INFORMATION

Toxicity	no data available
Persistence and degradability	no data available
Bioaccumulative potential	no data available
Mobility in soil	no data available
Results of PBT and vPvB assessment	PBT/vPvB assessment not available as chemical safety assessment not required/not conducted
Other adverse effects	no data available

13. DISPOSAL CONSIDERATIONS

Waste treatment methods	Offer surplus and non-recyclable solutions to a licensed disposal company. Contact a licensed professional waste disposal service to dispose of this material.
Contaminated packaging	Dispose of as unused product.

14. TRANSPORT INFORMATION

14.1 UN number

ADR/RID: UN 2924	IMDG: UN 2924	IATA-DGR: UN 2924
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14.2 UN proper shipping name

ADR/RID:	solid, corrosive, n.o.s (3,4-Dichloro-1-butene)
IMDG:	solid, corrosive, n.o.s (3,4-Dichloro-1-butene)
IATA:	solid, corrosive, n.o.s (3,4-Dichloro-1-butene)

14.3 Transport hazard class(es)

ADR/RID: 3(8)	IMDG: 3(8)	IATA-DGR: 3(8)
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14.4 Packaging group

ADR/RID: III	IMDG: III	IATA-DGR: III
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14.5 Environmental hazards

ADR/RID: -	IMDG: -	IATA-DGR: -
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14.6 Special precautions for user

Further information :	No data available
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15. REGULATORY INFORMATION

SARA 302:	No chemicals in this material are subject to the reporting requirements of SARA Title III, Section 302.
SARA 313:	This material does not contain any chemical components with known CAS numbers that exceed the threshold (De Minimis) reporting levels established by SARA Title III, Section 313.
SARA 311/312 Hazards	Acute Health Hazard

Massachusetts Right To Know Components	No components are subject to the Massachusetts Right to Know Act.
Pennsylvania Right To Know Components	
New Jersey Right To Know Components	
California Prop. 65 Components	This product does not contain any chemicals known to State of California to cause cancer, birth defects, or any other reproductive harm.

16. OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3.

Acute Tox.	Acute toxicity
Eye Irrit.	Eye irritation
Skin Irrit.	Skin irritation
H226	Flammable liquid and vapor
H314	Causes severe skin burns and eye damage
H402	Harmful to aquatic life
H302+H332	Harmful if swallowed or if inhaled

Further information

The above information is believed to be correct but does not purport to be all inclusive and shall be used only as a guide. The information in this document is based on the present state of our knowledge and is applicable to the product with regard to appropriate safety precautions. It does not represent any guarantee of the properties of the product. Angene shall not be held liable for any damage resulting from handling or from contact with the above product. See invoice or packing slip for additional terms and conditions of sale.